

TECHNICAL DRAWING SET

Network Flows

Network zones, flow matrix, web request path, host firewall and camera configuration.

Reference installation: Lenovo ThinkCentre M73 Tiny · Ubuntu Server 26.04 LTS · ONVIF PTZ camera · Novoferm Novomatic 200 garage door operator. 100 % on-premises processing.

DOCUMENT CONTROL

Document number	JRV-DWG-001
Revision	B
Software version	Jarvis 0.9.0 (jarvis/VERSION)
Date of issue	2026-09-30
Author	Thierry Gayet <thierry.gayet@labworks.fr>
License	0BSD (Zero-Clause BSD)
Language	US English

RECORD OF REVISIONS

Rev	Date	Description
A	2026-09-29	Initial release (French edition).
B	2026-09-30	US English edition. Content aligned with the current repository (web front end with Anubis, MQTT, remote syslog, monitoring, NTP, TLS certificate lifecycle, camera power and end-of-life data). Contents with page numbers, glossary, bibliography and index added.

CONTENTS

1 Scope 3

 1.1 Flow numbering 3

2 Network overview 4

3 Web request path and local channels 5

4 Flow matrix 6

5 Host firewall (nftables) 8

 5.1 Deployed ruleset 8

 5.2 Notes 8

6 Camera and host configuration 9

 6.1 Camera side 9

 6.2 Mini-PC and LAN side 9

Appendix A Glossary 10

Appendix B Bibliography 12

 B.1 Books in English 12

 B.2 Books in French 12

 B.3 Standards, specifications and papers 12

Appendix C Index 13

1 SCOPE

This document describes every network flow of the Jarvis smart gatekeeper: the network zones, the flow matrix, the web request path through nginx and Anubis, the host firewall and the configuration of the camera and of the mini-PC. All processing (vision, voice, decision) runs on the mini-PC: in operation, no flow goes to a cloud service. Internet access is only needed for the installation, the updates and the optional services marked as such.

NOTE

The addresses are examples. Report the actual values in `/etc/jarvis/config.yaml` (`camera.rtsp_url`, `ptz.host`) and in the Ansible inventory (`jarvis_lan_networks`, `jarvis_admin_networks`, `jarvis_monitoring_server`).

CAUTION

Never forward port 443 (or 80) from the Internet router to Jarvis. Remote access goes through a VPN only (F7). The Let's Encrypt DNS-01 challenge (F23) needs no inbound port; the HTTP-01 challenge would require port 80 reachable from the Internet.

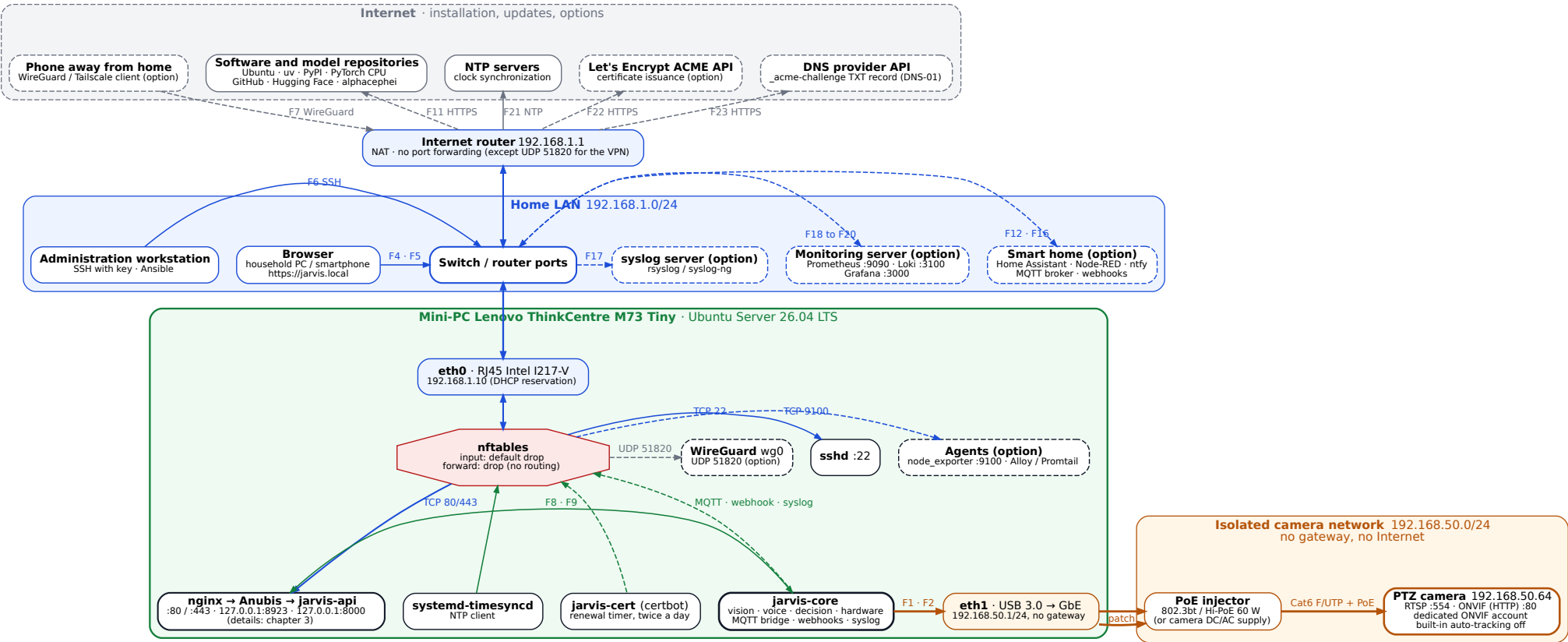
1.1 Flow numbering

Flows F1 to F12 keep the numbers of the previous revision. F13 to F24 cover the web front end (nginx, Anubis), the smart-home bridge (MQTT), the log and metrics exports (syslog, Loki, Prometheus, node_exporter), the time synchronization (NTP) and the TLS certificate lifecycle (ACME, DNS-01).

Group	Flows
Camera (eth1)	F1 RTSP, F2 ONVIF, F3 NTP (option)
Users and administration	F4 HTTPS, F5 HTTP, F6 SSH, F7 WireGuard (option), F24 mDNS
Inside the mini-PC	F8 control socket, F9 preview file, F10 database, F13 to F15 web front end
Integrations (LAN)	F12 webhooks, F16 MQTT, F17 syslog, F18 Loki, F19 metrics, F20 node_exporter
Internet	F11 installation and updates, F21 NTP, F22 ACME, F23 DNS-01

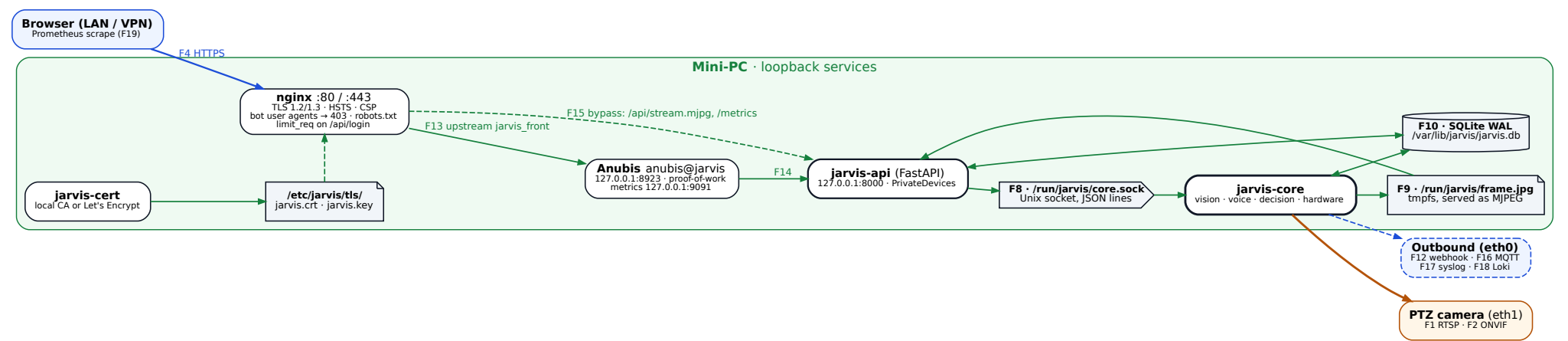
2 NETWORK OVERVIEW

home LAN isolated camera network inside the mini-PC Internet / option (dashed) filtering



3 WEB REQUEST PATH AND LOCAL CHANNELS

nginx terminates TLS and filters bots; Anubis (proof-of-work) protects the login page and the UI; the API only listens on the loopback interface and reaches the hardware through the closed command set of the control socket.



```
client → nginx :443 (TLS, headers, bot UA 403, robots.txt, limit_req on /api/login)
├─ upstream jarvis_front → Anubis 127.0.0.1:8923 → API 127.0.0.1:8000 → /run/jarvis/core.sock → jarvis-core
└─ /api/stream.mjpg, /metrics → upstream jarvis_api → API 127.0.0.1:8000
```

4 FLOW MATRIX

No.	Source	Destination	Protocol / port	Direction	Content	Indicative rate
F1	jarvis-core (eth1)	camera 192.168.50.64	RTSP over TCP 554	outbound, long session	Secondary H.264 stream 1280×720, 10 to 15 fps. Digest authentication.	≈ 1 to 2 Mbit/s in
F2	jarvis-core (eth1)	camera 192.168.50.64	ONVIF SOAP / HTTP TCP 80	outbound, on demand	PTZ ContinuousMove / Stop / GotoPreset, at most 4 commands/s, WS-Security.	< 20 kbit/s
F3	camera	mini-PC 192.168.50.1	NTP UDP 123	inbound (option)	Camera time without Internet access; needs an NTP server (e.g. chrony) on eth1, not deployed by default.	negligible
F4	browser (LAN / VPN)	nginx :443	HTTPS, TLS 1.2/1.3	inbound	Web UI, REST API, MJPEG stream of the Live tab, photo and voice uploads (16 MB max.).	≈ 2 to 5 Mbit/s per open Live tab
F5	browser (LAN)	nginx :80	HTTP	inbound	301 redirect to HTTPS; ACME HTTP-01 files under /.well-known/acme-challenge/ only.	negligible
F6	admin workstation	sshd :22	SSH	inbound	Administration and Ansible. Key authentication, admin networks only, 15 new connections/min per source.	—
F7	remote phone	router → mini-PC :51820	WireGuard UDP 51820	inbound (option)	Remote access. The only port that may be forwarded. With Tailscale: no forwarding at all.	as F4
F8	jarvis-api	jarvis-core	Unix socket /run/jarvis/core.sock	local	One JSON line per request, closed command list: status, reload_*, enroll_*, embed_face, garage_pulse, ptz_home, reload_settings, system, restart, say.	—
F9	jarvis-core	jarvis-api	file /run/jarvis/frame.jpg (tmpfs)	local	Annotated preview, replaced atomically. The RTSP stream is decoded only once.	5 fps
F10	jarvis-core and jarvis-api	SQLite WAL	file /var/lib/jarvis/jarvis.db	local	Persons, embeddings, unknown faces, sightings, log, accounts, sessions, settings.	—
F11	mini-PC (eth0)	Internet	HTTPS TCP 443 (+ DNS 53)	outbound, installation	apt, uv, PyPI, PyTorch CPU, models (GitHub, Hugging Face, alphacephei), security updates.	≈ 3 GB once
F12	jarvis-core (eth0)	Home Assistant / Node-RED / ntfy	HTTP(S) POST JSON	outbound (option)	Notifications (unknown visitor, watchlist, out-of-schedule access, voice denied, garage pulse), notifications.webhook_url.	a few KB per event
F13	nginx	Anubis 127.0.0.1:8923	HTTP, loopback	local	Upstream jarvis_front: every request except the bypass paths (F15). Without Anubis, the upstream points to the API directly.	as F4
F14	Anubis	jarvis-api 127.0.0.1:8000	HTTP, loopback	local	Requests that passed the bot policy (browser clients solve the proof-of-work once, cookie).	as F4
F15	nginx	jarvis-api 127.0.0.1:8000	HTTP, loopback	local	Upstream jarvis_api: /api/stream.mjpg (no buffering, session checked by the API) and /metrics (bearer token and source address checked by the API).	as F4

No.	Source	Destination	Protocol / port	Direction	Content	Indicative rate
F16	jarvis-core	MQTT broker (LAN)	MQTT TCP 1883, 8883 with TLS	outbound (option)	Retained status/state topics, event topics, Home Assistant discovery; inbound commands (garage_pulse, ptz_home, say) on the same connection only with <code>mqtt.allow_commands</code> and the command token.	a few KB per event
F17	jarvis-core, jarvis-api	rsyslog / syslog-ng server	syslog RFC 3164, UDP or TCP 514	outbound (option)	Copy of the application log (<code>logging.syslog_*</code>).	low
F18	Alloy / Promtail agent	Loki :3100	HTTP push /loki/api/v1/push	outbound (option)	<code>/var/log/jarvis/*.log</code> and the systemd journal (<code>monitoring.loki_url</code>).	low
F19	Prometheus (monitoring server)	nginx :443 /metrics	HTTPS, bearer token	inbound (option)	Jarvis metrics (vision, camera, sightings, sessions, audit, disk), every 30 s.	low
F20	Prometheus (monitoring server)	node_exporter :9100	HTTP	inbound (option)	Host metrics; the firewall accepts TCP 9100 from the configured monitoring server only.	low
F21	systemd-timesyncd	NTP servers	NTP UDP 123	outbound	Clock synchronization (certificates, audit timestamps).	negligible
F22	jarvis-cert (certbot)	Let's Encrypt ACME API	HTTPS TCP 443	outbound (option)	Issuance and renewal of the certificate in <code>letsencrypt</code> mode (renewal check twice a day).	negligible
F23	certbot DNS plugin	DNS provider API	HTTPS TCP 443	outbound (option)	DNS-01 challenge: TXT record <code>_acme-challenge</code> . No inbound port is opened on the Internet.	negligible
F24	LAN devices	mini-PC UDP 5353	mDNS (Avahi)	inbound	Name resolution of <code><hostname>.local</code> , LAN only (option <code>jarvis_mdns</code>).	negligible

NOTE

Why a second physical network? The Tiny has a single RJ45 port. A USB 3.0 to Gigabit adapter (Realtek RTL8153, `r8152` kernel driver, or ASIX AX88179) gives the camera a dedicated segment with no gateway and no DNS: the camera can neither reach the Internet nor be reached from the LAN, and the RTSP stream does not load the home Wi-Fi. **Alternative:** a managed switch with a camera VLAN (802.1Q) and an `eth0.50` interface on the mini-PC.

5 HOST FIREWALL (NFTABLES)

5.1 Deployed ruleset

```
table inet jarvis_filter {
  chain input {
    type filter hook input priority filter; policy drop;
    iif "lo" accept
    ct state invalid counter drop
    ct state established,related accept
    # ICMP / ICMPv6 essentials, echo rate-limited (10/s); DHCP / DHCPv6 client replies
    ip saddr @lan_v4 udp dport 5353 accept # F24 mDNS (jarvis_mdns)
    ip saddr @admin_v4 tcp dport 22 ct state new \
      add @ssh_meter_v4 { ip saddr limit rate 15/minute } accept # F6
    ip saddr @lan_v4 tcp dport { 80, 443 } accept # F4, F5
    ip saddr <monitoring server> tcp dport 9100 accept # F20 (option)
    # jarvis_firewall_extra_rules, e.g. F3 or F7:
    # iifname "eth1" ip saddr 192.168.50.64 udp dport 123 accept
    # udp dport 51820 accept
    limit rate 6/minute log prefix "jarvis-fw-drop: " level info
    counter drop comment "jarvis dropped"
  }
  chain forward { type filter hook forward priority filter; policy drop; } # no routing
  chain output { type filter hook output priority filter; policy accept; } # F1, F2, F11, F12, F16 to F18, F21 to F23
}
```

Simplified from `deploy/ansible/roles/firewall/templates/nftables.conf.j2` (IPv6 rules omitted). The ruleset is validated with `nft -c` before it replaces `/etc/nftables.conf`; only the `jarvis_filter` table is replaced, fail2ban keeps its own table.

5.2 Notes

- Input policy **default-deny**; SSH only from `jarvis_admin_networks`, HTTP/HTTPS only from `jarvis_lan_networks`, `node_exporter` only from `jarvis_monitoring_server`.
- Forwarding is dropped (`ip_forward = 0`): no routing between the LAN and the camera network.
- Output is accepted: camera RTSP/ONVIF, apt, NTP, MQTT, webhooks, syslog, Loki, ACME.

- fail2ban (nftables actions) adds the `sshd`, `jarvis-login`, `nginx-botsearch` and `recidive` jails.
- Anubis (127.0.0.1:8923, metrics 127.0.0.1:9091) and the API (127.0.0.1:8000) listen on the loopback interface only: no firewall rule exposes them.

CAUTION

The camera NTP flow (F3) and the WireGuard port (F7) are not accepted by the deployed ruleset. Add them through `jarvis_firewall_extra_rules` only when they are used.

6 CAMERA AND HOST CONFIGURATION

6.1 Camera side

- Fixed IP **192.168.50.64/24**, **no gateway**, empty DNS; NTP server 192.168.50.1 when F3 is used.
- Disable UPnP, vendor P2P / cloud services, Telnet/SSH and unused ports.
- Create a dedicated **ONVIF** user (PTZ and stream rights), distinct from the admin account.
- Secondary stream **1280×720 H.264**, 10 to 15 fps, constant bit rate 1 to 2 Mbit/s.
- Preset 1 = view of the entrance. Built-in auto-tracking **disabled**. Mask the public road.

6.2 Mini-PC and LAN side

- DHCP reservation (or static IP) for **eth0**; name `jarvis.local` through Avahi/mDNS.
- **eth1** with the static address 192.168.50.1/24 (netplan), no gateway.
- nginx listens on 80/443; Anubis and FastAPI stay on the loopback interface, never exposed.
- Certificate: `jarvis-cert` in `local` mode (private name-constrained CA, imported once on the clients) or `letsencrypt` mode (DNS-01 by default, F22 and F23).
- Docker variant: nginx publishes 80/443, the API stays on the internal Docker network, the core and the API share the `jarvis-data` and `jarvis-run` volumes.
- After the installation, F11 can be blocked: operation is fully offline (F21 remains advisable).

APPENDIX A GLOSSARY

Technical terms and abbreviations used in this document.

802.3af / at / bt

IEEE Power over Ethernet standards: 15.4 W (af, "PoE"), 30 W (at, "PoE+"), 60 to 90 W at the source (bt, "PoE++", classes 5 to 8). The PTZ cameras of the project draw up to 51 W and need 802.3bt or the vendor's Hi-PoE injector.

ACME

Automatic Certificate Management Environment (RFC 8555): protocol used by certbot to obtain certificates from Let's Encrypt, with an HTTP-01 or DNS-01 challenge.

Alloy / Promtail

Grafana agents that ship log files and the systemd journal to Loki. Grafana has deprecated Promtail in favor of Alloy.

Anubis

Proof-of-work reverse proxy that keeps crawlers, AI agents and scrapers away from the web UI (instance anubis@jarvis on 127.0.0.1:8923).

API (REST)

HTTP programming interface in which each resource (person, sighting...) has a URL and is handled with the GET, POST, PATCH and DELETE verbs.

CA (certificate authority)

Entity that signs certificates. Jarvis either runs a private, name-constrained "Jarvis Local CA" or uses Let's Encrypt.

certbot

ACME client of the EFF used by jarvis-cert to obtain and renew Let's Encrypt certificates.

CSP

Content Security Policy: HTTP header that restricts the sources of scripts, images and frames of a page.

DHCP

Automatic IP address assignment protocol; a "DHCP reservation" always gives the same address to a network card.

DNS-01

ACME challenge proven by publishing a TXT record in the public DNS zone through the DNS provider API. It needs no inbound port open on the Internet.

Embedding

Representation of a face or a voice as a numeric vector; two close embeddings probably belong to the same person.

F/UTP

Twisted-pair cable with an overall foil screen (Foiled / Unshielded Twisted Pairs).

fail2ban

Log-driven intrusion prevention: bans the source addresses that repeatedly fail SSH or web logins.

Grafana

Dashboard server of the optional monitoring stack (Loki and Prometheus data sources).

H.264 / H.265

Video compression standards (AVC / HEVC). The analyzed stream must be H.264, the only one the Intel HD 4600 decodes in hardware.

Hi-PoE / High PoE

Proprietary high-power PoE (Hikvision, Dahua; "High PoE" at Axis), close to 802.3bt.

Home Assistant

Open-source home automation controller; consumes the Jarvis MQTT discovery messages and webhooks.

HSTS

HTTP Strict Transport Security: forces the browser to use HTTPS for the site.

Injector (PoE midspan)

Unit that adds power to an Ethernet cable between a network device and a powered device.

Let's Encrypt

Free public certificate authority, reached with the ACME protocol.

Loki

Grafana log store; receives the Jarvis logs pushed by Alloy or Promtail (port 3100).

mDNS

Multicast DNS (Avahi): resolves <hostname>.local on the LAN without a DNS server.

MJPEG

Video stream made of a sequence of JPEG images, played directly by a browser.

MQTT

Lightweight publish/subscribe messaging protocol (TCP 1883, 8883 with TLS) used by the optional smart-home bridge.

NAT

Network Address Translation performed by the Internet router, which isolates the LAN from the Internet.

nftables

Linux kernel firewall (successor of iptables).

node_exporter

Prometheus agent that exposes the host metrics (CPU, memory, disk, network) on port 9100.

NTP

Network Time Protocol (UDP 123); keeps the clocks synchronized (certificates, audit timestamps).

ONVIF

IP camera interoperability standard (discovery, streaming, PTZ). Profile S: video and PTZ.

PoE

Power over Ethernet: electrical power carried by the network cable.

Preset

PTZ position stored in the camera (pan, tilt, zoom).

Prometheus

Metrics collection server; scrapes the Jarvis /metrics endpoint and node_exporter.

PTZ

Pan-Tilt-Zoom: camera motorized in pan, tilt and zoom.

Pulse

Short closure (500 ms) of the relay contact, equivalent to pressing the push button.

RTSP

Real Time Streaming Protocol (RFC 2326/7826): streaming protocol of the camera video.

Sighting

Time-stamped record of an identification: date, time, identity or unknown, score, photo.

SQLite (WAL)

Database embedded in a single file; Write-Ahead Log mode allows concurrent reads and a write.

syslog

Standard log transport (RFC 3164, UDP or TCP 514) to an rsyslog or syslog-ng server.

systemd

Linux service manager (jarvis-core, jarvis-api units, timers).

TLS / HTTPS

Encryption of web traffic (Transport Layer Security).

tmpfs

File system held in RAM (/run).

Unix socket

Local inter-process communication channel represented by a file (/run/jarvis/core.sock).

Unknown cluster

Automatic grouping of the visits of the same unknown visitor.

uv

Python package and environment manager (Astral).

VLAN

Virtual LAN (802.1Q) that logically separates devices on the same switch.

Watchlist

Flagged persons whose recognition triggers an alert and never grants access.

Webhook

HTTP request sent automatically to another service when an event occurs.

WireGuard

Modern, lightweight VPN protocol, recommended for remote access.

APPENDIX B BIBLIOGRAPHY

Further reading. Titles are given as published; ISBNs are those of the edition cited and are omitted when they could not be verified.

B.1 Books in English

1. Anderson, Ross. *Security Engineering: A Guide to Building Dependable Distributed Systems*. 3rd ed. Wiley, 2020. ISBN 978-1-119-64278-7.
2. DeJonghe, Derek. *NGINX Cookbook: Advanced Recipes for High-Performance Load Balancing*. 3rd ed. O'Reilly, 2024. ISBN 978-1-098-15843-9.
3. Geerling, Jeff. *Ansible for DevOps: Server and Configuration Management for Humans*. 2nd ed. Leanpub, 2020.
4. Kerrisk, Michael. *The Linux Programming Interface: A Linux and UNIX System Programming Handbook*. No Starch Press, 2010. ISBN 978-1-59327-220-3.
5. Pivotto, Julien; Brazil, Brian. *Prometheus: Up & Running: Infrastructure and Application Performance Monitoring*. 2nd ed. O'Reilly, 2023. ISBN 978-1-098-13114-2.
6. Pulver, Tim. *Hands-On Internet of Things with MQTT*. Packt, 2019. ISBN 978-1-78934-178-2.
7. Ristić, Ivan. *Bulletproof TLS and PKI: Understanding and Deploying SSL/TLS and PKI to Secure Servers and Web Applications*. 2nd ed. Feisty Duck, 2022. ISBN 978-1-907117-09-1.
8. Stevens, W. Richard; Fenner, Bill; Rudoff, Andrew M. *UNIX Network Programming, Volume 1: The Sockets Networking API*. 3rd ed. Addison-Wesley, 2003. ISBN 978-0-13-141155-5.
9. Ward, Brian. *How Linux Works: What Every Superuser Should Know*. 3rd ed. No Starch Press, 2021. ISBN 978-1-7185-0040-2.

B.2 Books in French

1. Blaess, Christophe. *Développement système sous Linux : ordonnancement multitâche, gestion mémoire, communications, programmation réseau*. 5th ed. Eyrolles, 2019. ISBN 978-2-212-67760-7.
2. Ghernaouti, Solange. *Cybersécurité : analyser les risques, mettre en œuvre les solutions*. 8th ed. Dunod, 2025. ISBN 978-2-10-087183-4.
3. Pujolle, Guy. *Les Réseaux*. 2024-2026 edition (10th ed.). Eyrolles, 2024. ISBN 978-2-416-01433-8.
4. Rohaut, Sébastien; Banquet, Philippe. *Linux : maîtrisez l'administration du système*. 7th ed. Éditions ENI, 2024. ISBN 978-2-409-04366-6.
5. Tanenbaum, Andrew S.; Wetherall, David J. *Réseaux*. 5th ed. Pearson, 2011. ISBN 978-2-7440-7521-6.

B.3 Standards, specifications and papers

1. ONVIF. *Profile S Specification; PTZ Service Specification*. onvif.org.
2. IETF RFC 7826. *Real-Time Streaming Protocol Version 2.0*, 2016 (RTSP 1.0: RFC 2326, 1998).
3. IETF RFC 8446. *The Transport Layer Security (TLS) Protocol Version 1.3*, 2018.
4. IETF RFC 8555. *Automatic Certificate Management Environment (ACME)*, 2019.
5. IETF RFC 3164. *The BSD syslog Protocol*, 2001.
6. IETF RFC 5905. *Network Time Protocol Version 4*, 2010.
7. OASIS. *MQTT Version 5.0*, OASIS Standard, 2019.
8. IEEE Std 802.3bt-2018. *Physical Layer and Management Parameters for Power over Ethernet over 4 pairs*.
9. IETF RFC 9106. *Argon2 Memory-Hard Function for Password Hashing and Proof-of-Work Applications*, 2021.
10. OWASP. *CycloneDX Bill of Materials Specification 1.6*, 2024.

APPENDIX C INDEX

Page numbers refer to the body of the document (chapters, not appendices).

#
802.3af / at / bt, 4
A
ACME, 3-4, 6-8
Alloy / Promtail, 4, 7
Anubis, 3-6, 8-9
API (REST), 4-9
C
CA (certificate authority), 5, 9
certbot, 4, 7
CSP, 5
D
DHCP, 4, 8-9
DNS-01, 3-4, 7, 9
E
Embedding, 6
F
F/UTP, 4
fail2ban, 8
G
Grafana, 4
H
H.264 / H.265, 6, 9
Hi-PoE / High PoE, 4

Home Assistant, 4, 6-7
HSTS, 5
I
Injector (PoE midspan), 4
J
jarvis-api, 4-7
jarvis-cert, 4-5, 7, 9
jarvis-core, 4-7
L
Lenovo ThinkCentre M73 Tiny, 4, 7
Let's Encrypt, 3-5, 7
Loki, 3-5, 7-8
M
mDNS, 3, 7-9
MJPEG, 5-6
MQTT, 3-5, 7-8
N
NAT, 4
nftables, 4, 8
nginx, 3-9
node_exporter, 3-4, 7-8
NTP, 3-4, 6-9
O
ONVIF, 3-6, 8-9
P
PoE, 4

Preset, 9
Prometheus, 3-5, 7
PTZ, 4-6, 9
Pulse, 6
R
RTSP, 3-8
S
Schedule (time slot), 6
Session / login, 5-8
Sighting, 6-7
SQLite (WAL), 5-6
syslog, 3-5, 7-8
systemd, 4, 7
T
TLS / HTTPS, 3-8
tmpfs, 5-6
U
Ubuntu Server, 4
Unix socket, 3, 5-6
USB 3.0 to Gigabit adapter, 3-9
uv, 4, 6
V
VLAN, 7
W
Watchlist, 6
Webhook, 3-6, 8
WireGuard, 3-4, 6, 8